THE FIRST CHERN CLASS IS CENTRAL AND COMMUTES WITH BIVARIANT CLASSES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The line bundle over an invertible module and the formulae for its zero section. Same chapter chow.tex, labels lemma-linebundle (proof line 6346) and lemma-linebundle-formulae (proof line 6348, claimed as S14-algebra-linebundle-formulae). Expresses the pullback of the cap with the first Chern class as minus the pushforward of the class along the zero section, and gives injectivity of the pullback along the bundle.
- The properties of a bivariant class. Same chapter chow.tex, definition of a bivariant class used at proof lines 6354-6364. Allows the bivariant class to be moved through the flat pullback and the pushforward along the zero section, which is the displayed computation.

Source reference. Original lemma 0B74: sources/stacks-chow/pinned-original-source.tex, source lines 5737–5737. Navigation pointer for \ref{lemma-linebundle} used inside the counted span (line 6346); the argument that consumes it is retained in the span.

Source reference. Original lemma 0F96: sources/stacks-chow/pinned-original-source.tex, source lines 5775–5775. Navigation pointer for \ref{lemma-linebundle-formulae} used inside the counted span (line 6348); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup}

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used inside the counted span (line 6333); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Then*

(1) *$c_1(\mathcal{L}) \in A^1(X)$ is in the center of $A^*(X)$ and*
(2) *if $f : X' \to X$ is locally of finite type and $c \in A^*(X' \to X)$, then $c \circ c_1(\mathcal{L}) = c_1(f^*\mathcal{L}) \circ c$.*

Proof. Of course (2) implies (1). Let $p : L \to X$ be as in Lemma 0B74 and let $o : X \to L$ be the zero section. Denote $p' : L' \to X'$ and $o' : X' \to L'$ their base changes. By Lemma 0F96 we have

$$p^*(c_1(\mathcal{L}) \cap \alpha) = -o_*\alpha \quad \text{and} \quad (p')^*(c_1(f^*\mathcal{L}) \cap \alpha') = -o'_*\alpha'$$

Since c is a bivariant class we have

$$\begin{aligned} (p')^*(c \cap c_1(\mathcal{L}) \cap \alpha) &= c \cap p^*(c_1(\mathcal{L}) \cap \alpha) \\ &= -c \cap o_*\alpha \\ &= -o'_*(c \cap \alpha) \\ &= (p')^*(c_1(f^*\mathcal{L}) \cap c \cap \alpha) \end{aligned}$$

Since $(p')^*$ is injective by one of the lemmas cited above we obtain $c \cap c_1(\mathcal{L}) \cap \alpha = c_1(f^*\mathcal{L}) \cap c \cap \alpha$. The same is true after any base change by $Y \to X$ locally of finite type and hence we have the equality of bivariant classes stated in (2). □